

Personalized AI assistants could either promote or hinder the emergence of collective intelligence

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Abstract

Advances in artificial intelligence (AI) have enabled the wide use of personalized assistance for cognitive tasks, which enhanced individual performance on various fields. However, the impact of AI assistance on collective intelligence remains poorly understood. AI assistance may undermine the emergence of collective intelligence by reducing the individual diversity or introducing structured biases into a population. Here, we develop a human-AI collective prediction model mediated by social learning based on incentive structure by extending the existing works on collective intelligence. Our model uncovers that emergence of collective intelligence is depend on how human and AI interact. When AI works as if it is omniscient and gives assistance regardless of users' interests, collective intelligence will be promoted. However, when AI works as a chat-bot and gives assistance only based on users' interests, collective intelligence could be hindered due to diversity loss and overconfidence on AI. We showed that, to promote collective intelligence, penalizing or incentivizing AI usage is not reliable. Instead, elaborating more diversity-aware incentives could be the key to promote and maintain the collective intelligence.

Introduction

Collective intelligence (CI) describes a phenomenon in which a group’s ability to solve problems, make decisions, or adapt to changing environments exceeds each member’s ability [1, 2, 3, 4, 5]. This phenomenon, also referred to as wisdom of crowds, has thus served as a central mechanism underlying the success of diverse organizations, ranging from animal groups [6, 7, 8] and human societies [2, 9] to machine-learning systems [10, 11]. Understanding the conditions that promote and maintain CI therefore remains a fundamental question across social sciences [12], evolutionary biology [13], and computer science [14]. Addressing this question has become particularly timely as artificial intelligence (AI) has begun to reshape how individuals acquire information, make decisions, and contribute to collective outcomes [15].

The recent emergence of generative AIs and large language models (LLMs) has changed the way individuals engage in various cognitive tasks, ranging from simple everyday tasks to more complex tasks, including scientific discovery [16] and creative writing [17]. In particular, while the use of generative AI models, such as ChatGPT, has been rapidly increasing [18], the potential long-term impact of personal AI assistants on human society, especially the CI, remains uncertain. Some argue that AI can promote CI by enhancing individual expertise [19], and opinion aggregation [20]. Conversely, empirical evidence indicate that the use of AIs can limit the diversity of opinions [17, 21, 22], a key requirement for the emergence of CI [23, 24, 25].

Our ability to predict the long-term impact of personal AI assistants on human CI is further hampered by the apparent discrepancy in the incentives that shape AI uses. On one hand, many industrial sectors are pushing towards increased AI usages, where the number of “tokens” used is often perceived as a proxy for productivity [26, 27]. On the other hand, many academic institutions appear to discourage, and even penalize, AI uses, reflecting concerns about plagiarism, misinformation, and limited independence [28, 29]. Such discrepancy raises a question about whether directly incentivizing or penalizing AI usage will benefit or harm human CI. These observations further highlight the importance of identifying a robust incentive structure that can promote CI in the presence of personal AI assistants. Such incentive structure should seek to retain the benefits of AIs and prevent potential risk, such as preserving the diversity of individual opinions.

To address these questions, we extend previous approaches for modeling CI, where each individual makes independent predictions for the outcome from partial observations of underlying causal factors [12, 13, 30]. In particular, we allow each individual to independently consult with personal AI assistants and revise their predictions accordingly. The resulting model allows us to investigate the robustness of human CIs under different assumptions about the AI model’s ability to make predictions as well as incentive structures that are associated with the prediction task. In doing so, we assess the impact of directly incentivizing or penalizing AI uses on CI and further suggest a balanced incentive structure that can maintain CI in the presence of AI.

66 Main

67 Model

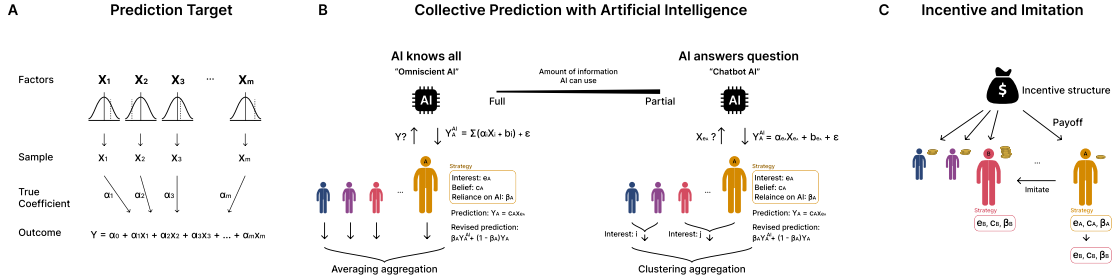


Figure 1: **A schematic diagram of the model for individual and collective prediction in the presence of personal AI assistants.** (A) The prediction target is modeled as a linear combination of causal factors. (B) Each player is characterized by three strategies: interest, belief, and reliance on AI. Players make predictions based on their interests and beliefs and subsequently revise them with AI assistance. We consider two types of AI, Chatbot AI and Omniscient AI, which differ in the amount of information available for generating predictions. (C) Players receive payoffs according to the incentive structure, and the population evolves through imitation of peers with higher payoffs.

68 We begin by extending previously established models of CI, in which a group of
 69 players participates in a prediction task (Figure 1): each player makes an individual
 70 prediction based on partial observations of the underlying causal factors, and the
 71 group forms a collective prediction by aggregating these individual predictions [30,
 72 12, 13]. Specifically, the prediction target (Y) is modeled as a linear combination of
 73 M causal factors x_i , with fixed coefficients α_i : $Y = \sum_{i=1}^M \alpha_i x_i + \alpha_0$, where each factor
 74 is drawn from independent normal distribution (Figure 1A). Then, each individual
 75 A observes a single factor of their interest x_{e_A} and makes a prediction based on
 76 their own belief c_A : $Y_A = c_A x_{e_A}$ (Figure 1B). We assume each individual can then
 77 independently consult their AI assistants and revise their prediction based on their
 78 reliance on AI β_A : $Y_A^{re} = \beta_A Y_A^{AI} + (1 - \beta_A) Y_A$ (Figure 1B). This revised prediction
 79 is aggregated to form collective prediction. **[SWP: I think we need to say averaging**
 80 **and clustering aggregation here?]**

81 To model AI, we assume that AI has already been trained on similar tasks and is
 82 able to make predictions based on partial or full observations of causal factors (Figure
 83 1B). Every time a player consults AI, it will make a new prediction Y_A^{AI} based on
 84 its internal parameters, which can include factor-specific bias (b_i) and stochastic
 85 error ϵ . Here, we consider two extreme types of AI response, which differ in the
 86 amount of information AI uses when making predictions. In one extreme, the AI
 87 can observe all causal factors and use all information to make predictions, regardless

88 of the user’s interest: $Y_A^{AI} = \sum_{i=0}^M (\alpha_i x_i + b_i) + \epsilon$. In the other extreme, AI works
 89 like a chatbot, where the AI predicts the outcome based only on the user’s interest:
 90 $Y_A^{AI} = \alpha_{e_A} x_{e_A} + b_{e_A} + \epsilon$. Hereafter, we refer to these two types as ‘Omniscient AI’
 91 and ‘Chatbot AI’, respectively.

92 Each generation consists of multiple round of predictions, and each player is re-
 93 warded a payoff according to the incentive structure (Figure 1C). For simplicity,
 94 we assume that a sufficiently large number of prediction rounds occurs within each
 95 generation, allowing us to use the expected payoff value rather than computing the
 96 payoffs for every round. We consider two incentive structures from previous work,
 97 “feedback” and “niche-expert”, which have been shown to promote near-perfect col-
 98 lective accuracy in the absence of AI [13]. The feedback structure gives higher payoff
 99 to individuals who can push the collective prediction toward the true target, regard-
 100 less of their individual predictive accuracy. [SWP: *maybe we need equations...*]
 101 In contrast, the niche-expert structure gives higher payoff to individuals who have
 102 both better predictive accuracy and a minority interest. [SWP: *maybe we need*
 103 *equations... but also say that this preserves diversity*] We note collective intelligence
 104 fails to emerge under the combination of averaging aggregation with niche-expert
 105 structure in the absence of AI [13].

106 Through time, the group evolves by allowing each individual to imitate peers
 107 who received higher payoffs (Figure 1C). Specifically, after each generation, two in-
 108 dividuals are randomly sampled, and one imitates all three strategies (the factor of
 109 interest, factor belief, and reliance on AI) of the other with a probability determined
 110 by the difference in their expected payoffs [SWP: *either refer to methods or supp.*].
 111 Individual predictions are aggregated into a collective prediction $\hat{Y}$ by simple averag-
 112 ing when using Omniscient AI and by clustering when using Chatbot AI. Collective
 113 accuracy is defined as the reduction in prediction error achieved by the collective
 114 prediction relative to a baseline prediction α_0 [SWP: *maybe need equation*]. Here-
 115 after, we refer to the near-perfect collective accuracy as accurate CI. [SWP: *How*
 116 *about: hereafter, we say that CI emerges when the collective accuracy reaches a near*
 117 *perfect level.*]

118 This model now allows us to explore the impact of AI on CI depending on the
 119 mode of AI and the incentive structure. Here, we begin by comparing the emergence
 120 of CI with and without AI and subsequently investigate a how different incentive
 121 structures affect the emergence CI in the presence of AI.

122 AI can either promote or prevent the emergence of CI

123 To understand the impact of AI, we first compare collective dynamics with and
 124 without AI. First, consistent with [13], simple imitation rules and incentive structures
 125 allow collective intelligence to emerge in the absence of AI (Fig. 2A,C,D). One
 126 exception is the use of simple averaging with niche-expert incentive structures, under
 127 which the collective accuracy of the population remains near zero [SWP: *because...*]
 128 (Fig. 2B) [SWP: *maybe need a supp figure*].

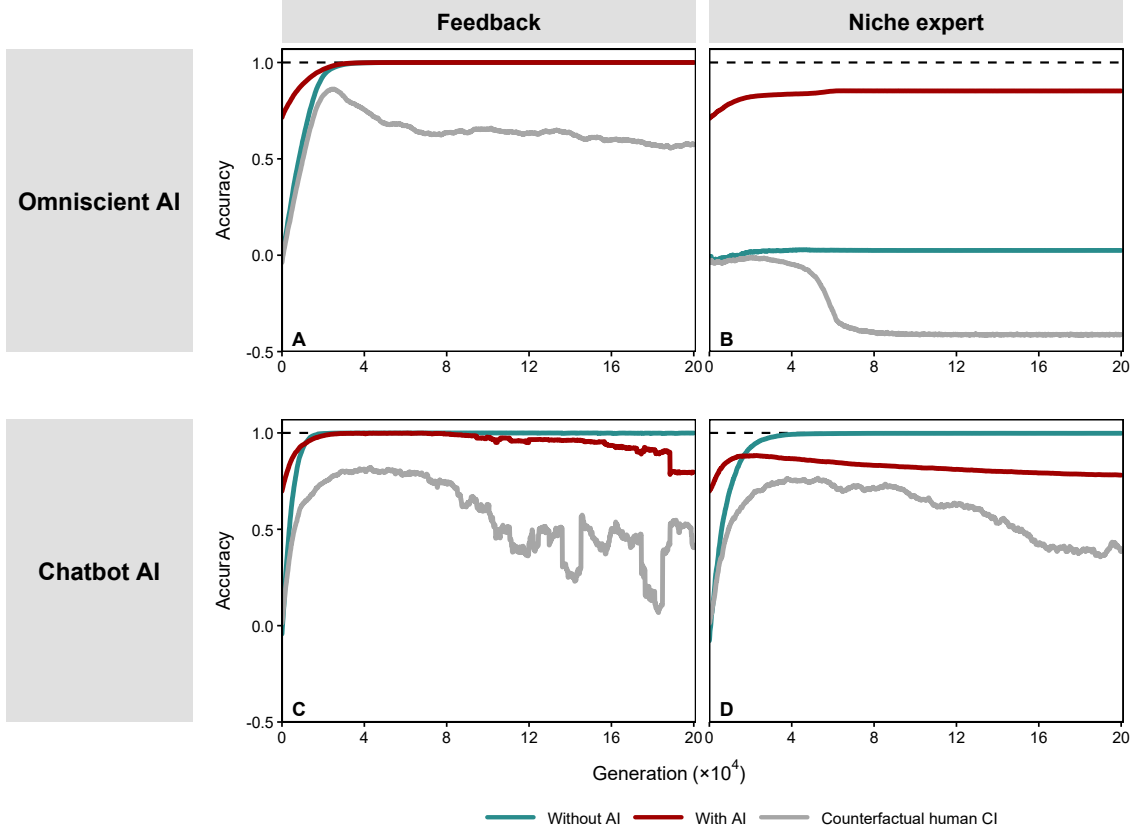


Figure 2: **Comparison of collective accuracy with and without AI.** Trajectories of collective accuracy when the group is assisted by the Omniscient AI (A,B) and Chatbot AI (C,D). [SWP: *Maybe we should also name the averaging rule below AI...*] Here, feedback (A,C) and niche-expert (B,D) incentive structures are considered. Green lines represent trajectories of collective accuracy without AI assistance. Red lines represent trajectories of collective accuracy with AI assistance. Gray lines represent trajectories of counterfactual collective accuracy, computed from players’ unrevised predictions, indicating the group’s predictive accuracy if AI assistance were unavailable at the given generation. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 100^2)$ for (A, B), $c_A \sim \mathcal{N}(0, 5^2)$ for (C, D), and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.36$ and $\tau = 0.3$.

129 We find that AI usage can have heterogeneous outcomes depending on its type.
130 For example, the use of Omniscient AI can either preserve CI under the feedback
131 incentive structure (Fig. 2A) or promote higher collective accuracy under the niche-
132 expert incentive structure (Fig. 2B). These results are robust across different levels
133 of AI accuracy and can be shown analytically by assuming a sufficiently large group
134 of players (Fig. S2) [SWP: *probably need to swap S2 and S1 so that the current S2*

135 *appears first*].

136 In contrast, the use of Chatbot AI prevents the emergence of CI, thereby resulting
137 in lower levels of collective accuracy (Fig. 2C, D). Under the feedback incentive
138 structure, the use of Chatbot AI leads to a loss of diversity in the population’s
139 interest, making it impossible to account for effect of some causal factors [SWP:
140 *Supp*]. This causes an increase in collective variance and stochastic decrease in
141 collective accuracy [SWP: *Supp*]. We note that the loss of interest diversity may
142 arise even in the absence of AI in finite populations [SWP: *Supp*], but the use of
143 Chatbot AI makes the group more vulnerable to such loss [SWP: *Supp*]. Under
144 the niche-expert incentive structure, the population becomes overly reliant on AI,
145 causing the collective prediction to inherit the AI’s bias (Fig. S1). This leads to
146 an increase in collective bias and a decrease in collective accuracy [SWP: *Supp*].
147 Interestingly, we find that the niche-expert structure can prevent the use of AI under
148 specific assumptions about the relationship between the true coefficient and AI bias
149 (Fig. S2). In this case, the population completely excludes AI assistance, allowing
150 the CI to emerge eventually. However, this phenomenon can substantially delay the
151 emergence of CI [SWP: *Supp*].

152 To investigate whether the group can retain its predictive ability when AI assis-
153 tance suddenly becomes unavailable at any given time, we quantify counterfactual
154 collective accuracy based on unrevised predictions. Across all scenarios, we find that
155 the counterfactual collective accuracy is lower than the collective accuracy of the
156 corresponding population.

157 In summary, an omniscient AI can promote CI, but AI operating with limited in-
158 formation may prevent the emergence CI. Moreover, reliance on AI can limit inherent
159 ability of the population to make accurate predictions on their own, raising concerns
160 about future AI usage. These findings further underscore the need for incentive
161 structures that can mitigate the potential risks associated with AI assistance.

162 **Directly incentivizing or penalizing AI use can additionally** 163 **harm CI**

164 Given the strong influence of AI on collective outcomes, a straightforward interven-
165 tion is to directly incentivize or penalize the use of AI. [SWP: *maybe a sentence*
166 *about how this reflects real world scenarios.*] To investigate whether such direct
167 incentives or penalties can help the emergence of CI, we modify the two incentive
168 structures by adding a payoff term proportional to individual reliance on AI:

$$\text{Feedback/Niche-expert} + \lambda\beta_A \quad (1)$$

169 where λ represent the strength of intervention ($\lambda > 0$) or penalty ($\lambda < 0$). [SWP:
170 *maybe we need to call incentive pi earlier on...*] [SWP: *also good to define beta*
171 *again here as a reminder.*]

172 Directly incentivizing or penalizing AI use can further reduce the collective ac-
173 curacy (Fig. 3C, E, G), except when Omniscient AI is used under the feedback

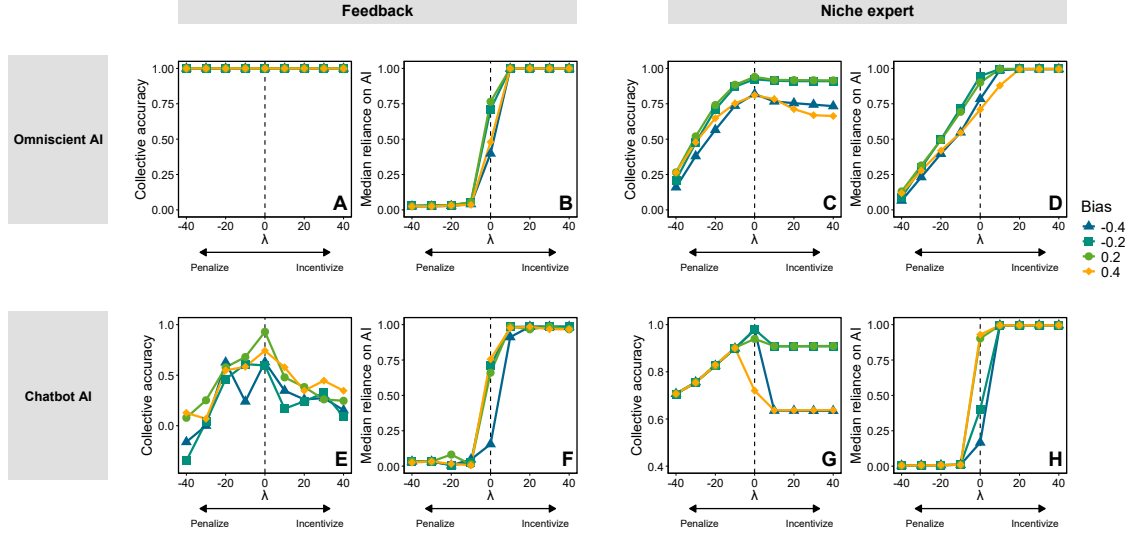


Figure 3: **Collective accuracy under the influence of direct intervention on AI use.** Collective accuracy and median reliance on AI under varying strengths of intervention (λ) across different assumptions about AI types (omniscient AI, A–D and chatbot AI, E–H) and incentive structures (Feedback, A,B,E,F, and Niche expert, C,D,G,H). Positive λ represents an incentive for AI use, whereas negative λ represents a penalty for AI use. In each panel, the vertical dashed line indicates the original incentive structure ($\lambda = 0$). Four AI bias levels are shown: $b_i = \pm 0.2$, representing higher-accuracy AI, and $b_i = \pm 0.4$, representing lower-accuracy AI. Collective accuracy (A,C,E,G) and median reliance on AI (B,D,F,H) are averaged over generations 190,000–200,000. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 100^2)$ for (A–D), $c_A \sim \mathcal{N}(0, 5^2)$ for (E–H), and $\beta_A \sim U(0, 1)$. AI parameter: $\tau = 0.3$.

structure; in the latter case, the population retains collective intelligence regardless of the intervention strength (Fig. 3A). The reduction in collective accuracy arises because incentivizing or penalizing AI usage causes individuals to imitate others based on their AI usage rather than their predictive accuracy. Thus, stronger interventions can push the group to always ($\beta = 1$) or never ($\beta = 0$) rely on AI (Fig. 3B,D, F, H) but these scenarios do not necessarily lead to better outcomes (Fig. 3D, F, H). When AI bias is relatively large, a weak penalty can lead to higher collective accuracy, but this represents a narrow parameter regime (Figure 3G).

Overall, these results indicate that simply incentivizing or penalizing the AI, independent of individuals’ performance, use may exacerbate rather than resolve the risk of AI on CI. Thus, we also considered an incentive structure where interventions for AI usage affects the payoffs multiplicatively, thus giving higher payoffs to individuals who rely more (or less) AI and makes better prediction. [SWP: Supp.]

Balanced incentives mitigate AI-induced failures of CI

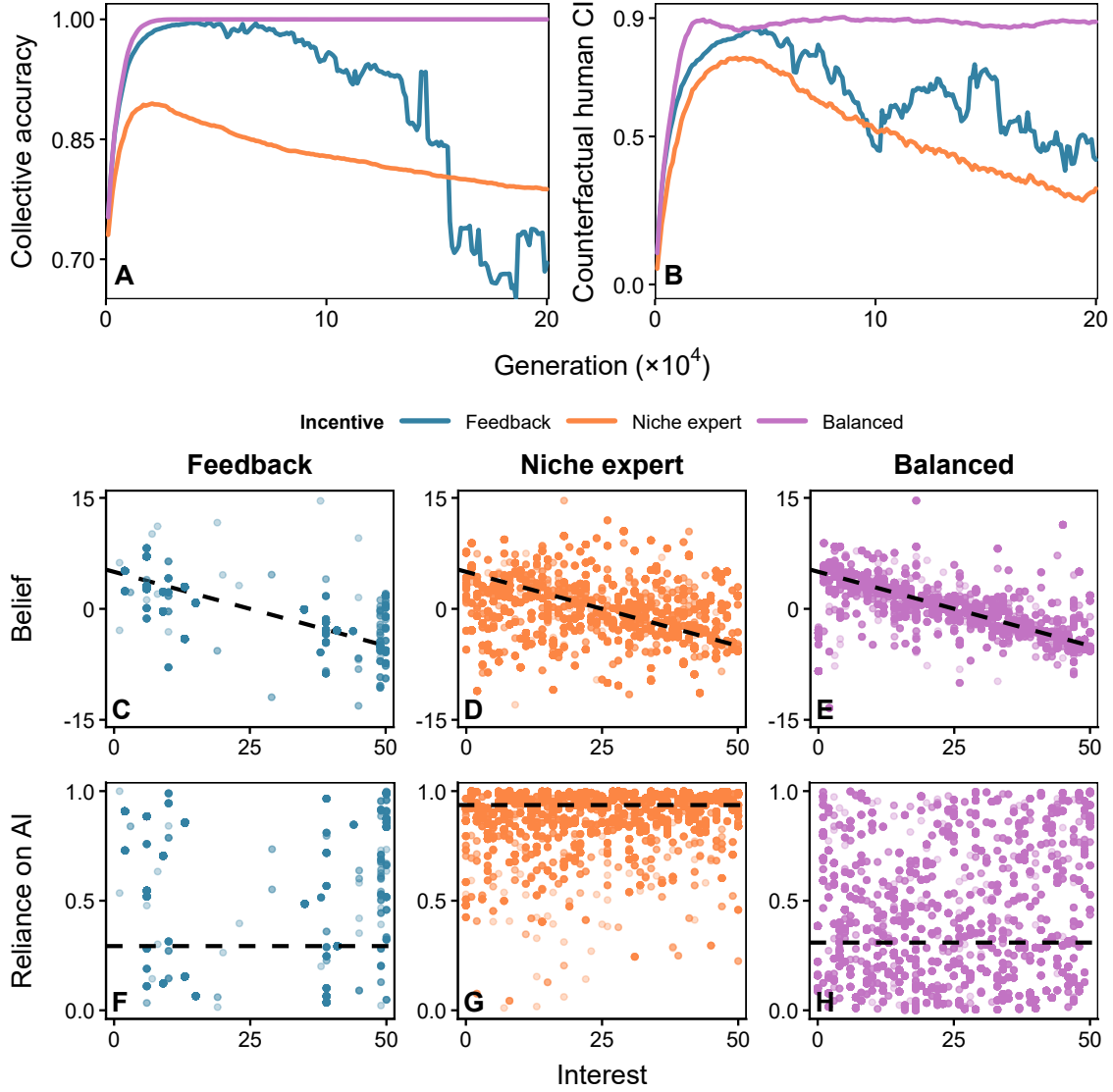


Figure 4: **Balanced structure can promote collective intelligence** Trajectories of collective accuracy (A) and counterfactual human CI (B) under three incentive structures (feedback, niche-expert, and balanced) with Chatbot AI. *[SWP: A and B should have the same y axes scaling.]* Distributions of individual interest and belief (C-E) and of interest and reliance on AI (F-H) after 200,000 generations. Each point represents an individual. Dashed lines in panels C-E represent the true coefficients (α_i) of the prediction target. Dashed horizontal lines in panels F-H indicate the median reliance on AI. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i = 5 - 0.2i$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 5^2)$, and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.4$ and $\tau = 0.3$.

188 Neither the original incentive structures nor simple interventions targeting AI
189 use are sufficient to promote CI. What incentive structure, then, would work in the
190 presence of AI? Here, we propose a “balanced” incentive structure, modeled as the
191 average of the feedback and niche expert incentives.

192 The core idea behind the balanced structure is to maintain interest diversity and
193 prevent collective bias. On one hand, the feedback incentive structure is prone to
194 the loss of interest diversity but can prevent collective bias by incentivizing players
195 whose predictions lie opposite to the collective prediction. On the other hand, the
196 niche-expert structure can maintain interest diversity because it directly incentivizes
197 minority interests. However the niche-expert structure cannot prevent collective bias
198 because, within the same factor of interest, individuals are rewarded only according
199 to their individual predictive performance, which in turn homogenize the predictions.

200 Therefore, when both incentive structures are combined, the population achieves
201 CI by maintaining both interest diversity and limited collective bias (Fig. 4, S3).
202 Compare to the feedback and niche-expert incentive structures, individuals are both
203 evenly distributed through factors (Fig. 4C–E), and their beliefs lie closer to true
204 values (Fig. 4F–H). Moreover, the balanced incentive structure allows for higher lev-
205 els of counterfactual collective accuracy compared to pure feedback and niche-expert
206 scenarios. These patterns are driven by the emergent property of the balanced struc-
207 ture, in which group increases its reliance on AI as the AI becomes more accurate,
208 demonstrating appropriately calibrated AI use.

209 Discussion

210 In this paper, we study how AI and incentive structures affect the emergence of CI.
211 Our model suggests that the effect of AI depends on the type of AI used by the
212 group. AI that can incorporate all relevant causal factors may improve group perfor-
213 mance, whereas AI that relies on users to gather information about the causal factors
214 may impair it. By examining different variants of incentive structures, we further
215 show that direct interventions targeting AI use generally harm group performance.
216 Instead, we suggest balanced structure which can promote CI while also generating
217 appropriately calibrated AI use as an emergent property.

218 Although Omniscient AI can produce group performance that is equal to or bet-
219 ter than that achieved without AI, such AI may be difficult to realize in practice
220 because it needs to be able to observe causal factors independently. Achieving this
221 capability would require hardwares, such as sensors, as well as autonomous data ac-
222 quisition capabilities. For this reason, most AI services currently available are better
223 represented by the Chatbot AI in our model, which highlights the potential risks of
224 AI use for CI.

225 In our model, individuals revise their predictions based on the AI’s assistance. It
226 is therefore unsurprising that AI assistance reduces the diversity of individual pre-
227 dictions, a pattern that has also been observed empirically. However, a reduction in

228 prediction diversity does not necessarily lead to the collapse of CI. The principal risk
 229 arises when the incentive structure fails to guide the group toward an appropriate
 230 set of strategies. The feedback structure leads to the breakdown of CI because it
 231 recruits more players to more important factors, which have larger coefficients. This
 232 characteristic creates a risk of interest-diversity loss in finite populations, a situa-
 233 tion in which no player remains interested in certain factors because most players
 234 shift toward relatively important factors. Without AI, this occurs when initial in-
 235 dividual predictions are not sufficiently diverse or when there is no source of novel
 236 ideas, such as mutation. Revising individual predictions with AI assistance exac-
 237 erbates this problem by pulling individuals’ predictions toward the AI’s prediction,
 238 thereby reducing prediction diversity even under the same belief distribution. The
 239 niche-expert structure hinders CI because it favors extreme levels of reliance on AI.
 240 Because individuals within each interest group are affected only by their own perfor-
 241 mance, incentives based on individual performance reach their maximum at either
 242 full reliance on AI or no reliance at all. This extreme reliance results in suboptimal
 243 collective accuracy or substantially delays the emergence of CI.

244 Because AI already generates substantial benefits as well as costs, organizations
 245 increasingly seek to improve collective performance by regulating AI use. Such in-
 246 terventions may take direct forms, including AI-use policies or token-based leader-
 247 boards, or may operate indirectly through other incentives, such as productivity
 248 rewards or the social stigma associated with AI use. However, our model suggests
 249 that directly incentivizing or penalizing AI use is unlikely to induce appropriately
 250 calibrated reliance on AI or to promote CI. Incentivizing AI use could lead to overre-
 251 liance, causing groups to propagate AI biases, whereas penalizing AI use could lead
 252 to underreliance, preventing groups from using AI even when it performs well in the
 253 relevant domain. Our result instead shows that appropriate AI use and accurate CI
 254 can be promoted by focusing on the classic properties of groups that have been em-
 255 phasized for decades—maintaining diversity and preventing collective bias—rather
 256 than on AI use itself.

257 The consistently limited counterfactual collective accuracy observed across AI
 258 types and incentive structures suggests that the presence of AI can degrade hu-
 259 man group’s own performance, consistent with empirical evidence. This may reduce
 260 collective performance when AI assistance becomes unavailable because of resource
 261 constraints, system failure, or model collapse, forcing the group to rebuild CI through
 262 further adaptation. The higher counterfactual collective accuracy achieved under the
 263 balanced structure indicates that an appropriate incentive structure can mitigate the
 264 risk of CI breakdown following the loss of AI assistance.

265 Our model and analysis have several limitations. First, the model considers only
 266 social learning among humans and treats the influence of AI as unidirectional. In
 267 reality, humans can learn from AI, while AI systems can also be further trained
 268 on human-generated data. Such bidirectional adaptation may further reduce group
 269 performance by increasing the similarity between human and AI predictions and
 270 thereby weakening their complementarity. Empirical studies of human–AI collabo-

ration likewise suggest that stronger correlations between human and AI predictions can diminish the benefits of collaboration. Second, we model AI errors using factor-specific additive biases and random error. Previous studies of AI bias and error suggest that additive biases may arise from skewed training data, or systematic calibration. However, AI errors may also exhibit more complex structures, including biased estimates of factor effects and correlations across prediction errors. Further empirical research is therefore needed to characterize the structure of AI bias and error, together with modeling approaches that can incorporate these more complex patterns. Finally, to retain the tractability of established CI models, we assume that the prediction target is a linear combination of causal factors, which does not hold for many real-world tasks. In particular, many machine learning methods used in AI systems are capable of capturing nonlinear relationships. Future research should therefore investigate how nonlinear relationships between causal factors and prediction targets affect the interaction between AI assistance and the emergence of CI.

Methods

Prediction target

Following previous models of CI, we model the prediction target as a linear combination of M independent causal factors:

$$Y = \alpha_0 + \alpha_1 X_1 + \cdots + \alpha_M X_M, \quad (2)$$

where the fixed coefficient α_i determines the magnitude and direction of the effect of factor i , and α_0 is a global intercept. Each causal factor X_i is modeled as an independent Gaussian random variable with mean zero and standard deviation σ_i .

Players

We consider a well-mixed population $\mathcal{N}$ consisting of N players engaged in a novel prediction task. In the absence of AI assistance, the strategy of player A is characterized by two variables: interest e_A and belief c_A . Player A observes the causal factor X_{e_A} and predicts the target by multiplying the observed factor value by their belief:

$$Y_A = c_A X_{e_A}, \quad (3)$$

where e_A can take an integer value from 0 to M . When $e_A = 0$, the player does not observe any causal factor and instead makes a constant prediction based solely on their belief. We refer to this as a stubborn strategy because the player's prediction does not respond to the realized values of the causal factors. For notational convenience, we assume that $X_0 = 1$.

When the group is assisted by AI, each player’s strategy additionally includes reliance on AI, denoted by β_A . We assume that each player consults the AI independently and revises their prediction by taking a weighted average of their own prediction and the AI-generated prediction:

$$Y_A^{\text{re}} = \beta_A Y_A^{\text{AI}} + (1 - \beta_A) Y_A, \quad (4)$$

where β_A ranges from 0 to 1. A value of $\beta_A = 1$ indicates complete reliance on AI, whereas $\beta_A = 0$ indicates that the player does not use the AI prediction.

AI models

We assume that the AI model has been trained on similar tasks and is therefore capable of making predictions based on causal factors. However, because the model is imperfect, its outputs are subject to systematic additive bias and stochastic error. The additive bias associated with factor i is denoted by b_i , and the stochastic error is denoted by ϵ . The AI is neither trained nor updated during the simulation, and its biases therefore remain fixed throughout the simulation. Although the bias is indexed by factor, all b_i , including b_0 , are assigned the same value in the main analyses for simplicity. The model formulation nevertheless allows biases to differ among factors.

We assume that the stochastic error follows a Gaussian distribution with mean zero and standard deviation τ . Each time a player consults the AI, an independent error is generated. Errors are therefore independent across players and across repeated consultations by the same player. Consequently, stochastic AI errors can be attenuated when independently generated predictions are aggregated across players. We therefore held τ constant and varied the magnitude of the additive bias to manipulate the level of AI accuracy across analyses.

We consider two extreme types of AI that differ in the amount of information available for prediction. The first is termed the Omniscient AI, which can observe all M causal factors and incorporate them into its prediction regardless of the user’s interest. The prediction generated by the Omniscient AI for player A is expressed as follows:

$$Y_A^{\text{AI}} = \alpha_0 + b_0 + \alpha_1 X_1 + b_1 + \cdots + \alpha_M X_M + b_M + \epsilon. \quad (5)$$

The second is termed the Chatbot AI, which can use only the single causal factor X_{e_A} associated with the player’s interest. The prediction generated by the Chatbot AI for player A is expressed as follows:

$$Y_A^{\text{AI}} = \alpha_{e_A} X_{e_A} + b_{e_A} + \epsilon. \quad (6)$$

Aggregation rules

The revised individual predictions are aggregated to form a collective prediction. We use two aggregation rules proposed in previous research, depending on the type of

336 AI used by the group.

337 When the group is assisted by the Omniscient AI, the AI component of each
 338 revised prediction contains information about all M factors, even though the indi-
 339 vidual’s own prediction is based on only one factor. Each revised prediction can
 340 therefore be treated as an estimate of the entire target. Accordingly, we calculate
 341 the collective prediction by taking the simple average of the revised individual pre-
 342 dictions:

$$\hat{Y}^{\text{re}} = \frac{Y_1^{\text{re}} + Y_2^{\text{re}} + \dots + Y_n^{\text{re}}}{N}. \quad (7)$$

343 When the group is assisted by the Chatbot AI, both the player’s initial prediction
 344 and the AI’s prediction are based on the same causal factor. Each revised predic-
 345 tion therefore contains information about only one factor and can be treated as an
 346 estimate of that factor’s contribution to the target. We consequently use clustering
 347 aggregation.

348 Clustering aggregation is performed by grouping individuals according to their
 349 interests and calculating the mean prediction within each group. The collective
 350 prediction is then obtained by summing the mean predictions of all nonempty groups:

$$\hat{Y}^{\text{re}} = \sum_{i:n_i>0} \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} Y_A^{\text{re}}, \quad (8)$$

351 where $\mathcal{A}_i$ is the set of players interested in factor i , and $n_i = |\mathcal{A}_i|$ is the number
 352 of players in that set.

353 When the group is not assisted by AI, collective predictions are computed from the
 354 unrevised individual predictions throughout the simulation. Both simple averaging
 355 and clustering can be used in the absence of AI. For fair comparison, we use the
 356 same aggregation rule as in the corresponding AI-assisted condition.

357 Incentive structures

358 Players receive payoffs determined by an incentive structure based on their strategies.
 359 We consider five incentive structures: feedback structure, niche-expert structure,
 360 direct intervention on AI use of these two structures, and a balanced structure.

361 The feedback structure does not directly reward individuals according to their
 362 own prediction accuracy. Instead, it rewards predictions that are aligned with the
 363 residual error of the collective prediction and therefore have the potential to move
 364 the collective prediction toward the true value. Given player A ’s revised prediction
 365 Y_A^{re} and the collective prediction $\hat{Y}^{\text{re}}$, the payoff is expressed as

$$\pi_A = Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}). \quad (9)$$

366 The niche-expert structure assigns higher payoffs to individuals who make accu-
 367 rate predictions while having an underrepresented interest. Specifically, the squared

368 prediction error of player A is weighted by $\rho_{e_A} = n_{e_A}/n$, the proportion of players
 369 interested in the same factor:

$$\pi_A = -\rho_{e_A}(Y_A^{\text{re}} - Y)^2. \quad (10)$$

370 Previous research has shown that these two incentive structures can produce
 371 nearly perfect CI through social learning, except when the niche-expert incentive is
 372 combined with simple averaging. They are therefore used as baseline structures for
 373 examining how AI assistance affects the emergence of CI.

374 To examine the effects of directly encouraging or discouraging AI use, we mod-
 375 ify the feedback and niche-expert structures by adding a term proportional to the
 376 player’s reliance on AI:

$$\pi_A = Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) + \lambda\beta_A. \quad (11)$$

$$\pi_A = -\rho_{e_A}(Y_A^{\text{re}} - Y)^2 + \lambda\beta_A. \quad (12)$$

377 Here, λ is a weighting coefficient representing the strength of the direct incentive
 378 or penalty applied to AI use. A positive value of λ rewards AI reliance, whereas a
 379 negative value penalizes it. When $\lambda = 0$, no direct incentive or penalty is applied,
 380 and the payoff functions reduce to the original feedback and niche-expert structures.

381 The balanced structure is defined as the arithmetic mean of the feedback and
 382 niche-expert payoff components:

$$\pi_A = \frac{Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) - \rho_{e_A}(Y_A^{\text{re}} - Y)^2}{2}. \quad (13)$$

383 Simulations

384 All simulations were conducted using an agent-based model with 50 causal factors
 385 ($M = 50$) and 10,000 players ($n = 10,000$). The parameters defining the prediction
 386 target were sampled at the beginning of the simulations and subsequently held fixed.

387 The coefficients α_i were sampled from a uniform distribution on $[-5, 5]$, except in
 388 the simulation presented in Figure 4, in which α_i was set to $5 - 0.2i$ for visualization.
 389 The standard deviations σ_i were sampled from a uniform distribution on $[0, 3]$.

390 Players’ initial strategies were also sampled from predefined probability distri-
 391 butions. The distribution of individual belief c_A depended on the aggregation rule.
 392 Under simple averaging, each factor’s contribution is scaled by the proportion of
 393 players interested in that factor. We therefore sampled beliefs from a Gaussian dis-
 394 tribution with mean zero and standard deviation 100, following the scaling used
 395 in previous work [13]. Under clustering aggregation, beliefs were sampled from a
 396 Gaussian distribution with mean zero and standard deviation 5. Because predictions
 397 are averaged separately within each interest group and then summed, no additional
 398 scaling of c_A is required. The initial interest of each player was sampled uniformly

399 from the integers between 0 and 50, and reliance on AI was sampled from a uniform
400 distribution on $[0, 1]$.

401 We assumed that prediction events occurred on a substantially faster timescale
402 than social learning. At each generation, we therefore calculated the expected payoff
403 $\mathbb{E}[\pi_A]$ of each player analytically rather than explicitly sampling prediction events.
404 The expectation was taken over the distributions of the causal factors $X_1, \dots, X_M$
405 and the stochastic AI error ϵ , conditional on the current population strategies and
406 the fixed target parameters. Closed form expression of expected payoffs were derived
407 for all incentive structures and are presented in the Supplementary Methods.

408 Each generation consisted of an expected payoff calculation followed by one op-
409 portunity for pairwise social learning. Two distinct players, A and B , were randomly
410 sampled, and player A imitated all three strategies of player B with probability

$$\frac{1}{1 + \exp(s(\mathbb{E}[\pi_A] - \mathbb{E}[\pi_B]))}, \quad (14)$$

411 where s represents the strength of social learning and was set to 50 in all sim-
412 ulations. We introduced a mutation rate μ , such that with probability μ , player A
413 did not imitate player B but instead redrew all strategy components from the same
414 distributions used to initialize the population. The mutation rate was set to zero in
415 all simulations except those presented in Supplementary Figure 4.

416 Unless otherwise specified, each simulation was run for 200,000 generations. When
417 a result was represented by a single summary value rather than as a temporal tra-
418 jectory, the metric was averaged over the final 10,000 generations.

419 To evaluate the dynamics of collective prediction, we defined six summary met-
420 rics: collective accuracy, collective bias, collective variance, counterfactual collective
421 accuracy, interest diversity, and median reliance on AI. At each population state,
422 expectations and variances were evaluated analytically over the distributions of the
423 causal factors and AI errors rather than estimated from sampled prediction events.
424 Collective bias and collective variance constitute the bias–variance decomposition of
425 the group level collective prediction error.

426 Collective accuracy was normalized relative to the baseline prediction α_0 . A neg-
427 ative value indicates that the collective prediction performs worse than the baseline.
428 Counterfactual collective accuracy was evaluated by setting $\beta_A = 0$ while retaining
429 the evolved interests and beliefs of the AI-assisted population. In calculating interest
430 diversity, we used the convention that $\rho_i \ln \rho_i = 0$ when $\rho_i = 0$ (Table 1).

431 Data and code availability

432 All data are stored in a publicly available GitHub repository ([https://github.com/](https://github.com/parklab-snu/collective-intelligence-with-AI)
433 [parklab-snu/collective-intelligence-with-AI](https://github.com/parklab-snu/collective-intelligence-with-AI)).

Table 1: Evaluation metrics used to analyze collective prediction dynamics.

Metric	Definition	Interpretation
Collective accuracy	$1 - \frac{\mathbb{E}[(Y - \hat{Y}^{\text{re}})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$	Accuracy of the revised collective prediction relative to the baseline prediction α_0 .
Collective bias	$\left(\mathbb{E}[Y - \hat{Y}^{\text{re}}]\right)^2$	Squared expected deviation of the revised collective prediction from the true target.
Collective variance	$\text{Var}(Y - \hat{Y}^{\text{re}})$	Variability of the revised collective prediction error across prediction events.
Counterfactual collective accuracy	$1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$	Accuracy obtained after removing AI use from the current population while retaining its evolved interests and beliefs.
Interest diversity	$\exp\left(-\sum_{i=0}^M \rho_i \ln \rho_i\right)$	Effective number of interests represented in the population.
Median reliance on AI	$\text{median}_{A \in N}(\beta_A)$	Median reliance on AI in the population.

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438 **Contributions**

439 Jaeseung Kim: Conceptualization, Formal Analysis, Investigation, Methodology,
440 Software, Visualization, Writing - Original Draft Preparation, Writing - Review &
441 Editing. Sang Woo Park: Conceptualization, Investigation, Methodology, Supervi-
442 sion, Writing - Original Draft Preparation, Writing - Review & Editing.

443 **Competing Interests**

444 The authors declare no competing interests.

445 Supplementary Text

446 1 Omniscient AI

447 Omniscient AI assumes that AI can incorporate all causal factors regardless of the
448 player's interest.

449 1.1 Model

$$Y_A^{AI} = \alpha_0 + b_0 + \alpha_1 X_1 + b_1 + \cdots + \alpha_M X_M + b_M + \epsilon. \quad (S1)$$

450 For simplicity, prediction of Omniscient AI can be written as:

$$Y^{AI} = Y + B + \epsilon \quad (S2)$$

451 where Y stands for true prediction target ($Y = \alpha_0 + \alpha_1 X_1 + \cdots + \alpha_M X_M$) and
452 B is sum of all fixed biases ($B = b_0 + b_1 + \cdots + b_M$), ϵ follows normal distribution
453 with zero mean, τ standard deviation with no correlation within or between players.
454 Player A's prediction and revised prediction can be written as follows:

$$Y_A = c_A X_{e_A} \quad (S3)$$

455

$$Y_A^{re} = \beta_A Y^{AI} + (1 - \beta_A) Y_A \quad (S4)$$

456 where c_A stands for player A's belief on the factor, e_A stands for the interest, and
457 β_A stands for reliance on AI.

458 1.2 Aggregation

459 Because Y_A^{AI} is subject to all M factors, we use simple averaging aggregation for
460 collective prediction.

$$\begin{aligned} \hat{Y}^{re} &= \frac{1}{N} \sum_{j=1}^N Y_j^{re} \\ &= \frac{1}{N} \sum_{j=1}^N \{ \beta_j Y^{AI} + (1 - \beta_j) Y_j \} \\ &= \frac{1}{N} \sum_{j=1}^N \{ \beta_j (Y + B + \epsilon) + (1 - \beta_j) c_j X_{e_j} \} \end{aligned}$$

461 Reparameterization for simple derivation and interpretability.

$$\bar{\beta} := \frac{1}{N} \sum_{j=1}^N \beta_j \quad (S5)$$

462

$$\bar{\eta} := \frac{1}{N} \sum_{j=1}^N \beta_j \epsilon \quad (\text{S6})$$

463

$$\gamma_i := \frac{1}{N} \sum_{A \in \mathcal{A}_i} (1 - \beta_A) c_A \quad (\text{S7})$$

464 $\bar{\beta}$ can be interpreted as mean reliance on AI, γ stands for the coefficient for the factor
 465 i which is solely based on players' belief.

466 By this reparameterization, we can write $\hat{Y}^{re}$ as follows.

$$\hat{Y}^{re} = \bar{\beta}(Y + B) + \sum_{i=0}^M \gamma_i X_i + \bar{\eta} \quad (\text{S8})$$

467 1.3 Expected payoffs

468 Feedback

$$\delta_0 := \alpha_0 - (\bar{\beta}(\alpha_0 + B) + \gamma_0) \quad (\text{S9})$$

469

$$\delta_i := \alpha_i - (\bar{\beta}\alpha_i + \gamma_i), \quad (i \neq 0) \quad (\text{S10})$$

470

$$Y - \hat{Y}^{re} = \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta} \quad (\text{S11})$$

471 See equation Eq. (S6) for the $\bar{\eta}$.

472 δ_0 is constant term and δ_i is independent of α_k ($k \neq i$), ϵ , and $\bar{\eta}$.

$$\begin{aligned} \mathbb{E}(\pi_A) &= \mathbb{E}[Y_A^{re}(Y - \hat{Y}^{re})] \\ &= \mathbb{E} \left[\left\{ \beta_A \alpha_0 + \beta_A \sum_{i=1}^M \alpha_i X_i + \beta_A B + \beta_A \epsilon + (1 - \beta_A) c_A X_{e_A} \right\} \left\{ \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta} \right\} \right] \\ &= \beta_A \alpha_0 \delta_0 + \beta_A \sum_{i=1}^M \delta_i \alpha_i \sigma_i^2 + \beta_A B \delta_0 - \frac{1}{N} \beta_A^2 \tau^2 + (1 - \beta_A) c_A \delta_{e_A} \sigma_{e_A}^2 \\ &= \beta_A \left\{ (\alpha_0 + B) \delta_0 + \sum_{i=1}^M \alpha_i \delta_i \sigma_i^2 \right\} + (1 - \beta_A) c_A \delta_{e_A} \sigma_{e_A}^2 - \frac{\beta_A^2 \tau^2}{N} \end{aligned}$$

473 **Niche-expert**

474 If $e_A = 0$, player does not observe causal factor and their unrevised prediction is same
 475 with their belief c_A . Such that, the expected payoff is

$$\begin{aligned}
 \mathbb{E}(\pi_A) &= -\mathbb{E}[\rho_0(Y_A^{re} - Y)^2] \\
 &= -\mathbb{E}[\rho_0\{(\beta_A - 1)Y + \beta_A B + \beta_A \epsilon + (1 - \beta_A)c_A\}^2] \\
 &= -\mathbb{E}\left[\rho_0\left\{(\beta_A - 1)(\alpha_0 - c_A) + \beta_A B + (\beta_A - 1)\sum_{i=1}^M \alpha_i X_i + \beta_A \epsilon\right\}^2\right] \\
 &= -\rho_0\left\{(1 - \beta_A)^2(\alpha_0 - c_A)^2 - 2\beta_A(1 - \beta_A)(\alpha_0 - c_A)B + \beta_A^2 B^2\right. \\
 &\quad \left.+ (1 - \beta_A)^2 \sum_{i \neq 0} \alpha_i^2 \sigma_i^2 + \beta_A^2 \tau^2\right\}
 \end{aligned}$$

476 Else,

$$\begin{aligned}
 \mathbb{E}(\pi_A) &= -\mathbb{E}[\rho_{e_A}(Y_A^{re} - Y)^2] \\
 &= -\mathbb{E}[\rho_{e_A}\{(\beta_A - 1)Y + \beta_A B + \beta_A \epsilon + (1 - \beta_A)c_A X_{e_A}\}^2] \\
 &= -\mathbb{E}\left[\rho_{e_A}\left\{(\beta_A - 1)\alpha_0 + \beta_A B + (\beta_A - 1)(\alpha_{e_A} - c_A)X_{e_A} + (\beta_A - 1)\sum_{i \neq e_A, 0} \alpha_i X_i + \beta_A \epsilon\right\}^2\right] \\
 &= -\rho_{e_A}\left\{(1 - \beta_A)^2 \alpha_0^2 - 2\beta_A(1 - \beta_A)\alpha_0 B + \beta_A^2 B^2 + (1 - \beta_A)^2(\alpha_{e_A} - c_A)^2 \sigma_{e_A}^2\right. \\
 &\quad \left.+ (1 - \beta_A)^2 \sum_{i \neq e_A, 0} \alpha_i^2 \sigma_i^2 + \beta_A^2 \tau^2\right\}
 \end{aligned}$$

477 **Incentivizing or penalizing AI use**

478 Because additive term $\lambda\beta$ is not a random variable, we can simply derive the expected
 479 payoff as

$$\mathbb{E}(\pi_{A, Feedback}) = \mathbb{E}[Y_A^{re}(Y - \hat{Y}^{re}) + \lambda\beta_A] = \mathbb{E}[Y_A^{re}(Y - \hat{Y}^{re})] + \lambda\beta_A. \quad (S12)$$

$$\mathbb{E}(\pi_{A, Niche-expert}) = -\mathbb{E}[\rho_{e_A}(Y_A^{re} - Y)^2 + \lambda\beta_A] = -\mathbb{E}[\rho_{e_A}(Y_A^{re} - Y)^2] + \lambda\beta_A. \quad (S13)$$

480 **1.4 Expected accuracy**

$$\mathbb{E}[(Y - \alpha_0)^2] = \mathbb{E}\left[\left(\alpha_0 + \sum_{i=1}^M \alpha_i X_i - \alpha_0\right)^2\right] = \sum_{i=1}^M \alpha_i^2 \sigma_i^2 \quad (S14)$$

481

$$\mathbb{E}[(Y - \hat{Y}^{re})^2] = \mathbb{E}\left[\left(\delta_0 + \sum_{i=1}^M \delta_i X_i - \hat{\eta}\right)^2\right] = \delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \frac{\tau^2}{N^2} \sum_{j=1}^N \beta_j^2 \quad (\text{S15})$$

482

$$\begin{aligned} R &= 1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]} \\ &= 1 - \frac{\delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \frac{\tau^2}{N^2} \sum_{j=1}^N \beta_j^2}{\sum_{i=1}^M \alpha_i^2 \sigma_i^2} \end{aligned}$$

483 2 Chatbot AI

484 2.1 Model

$$Y_A^{AI} = \alpha_{e_A} X_{e_A} + b_{e_A} + \epsilon \quad (\text{S16})$$

485 b_{e_A} stands for factor specific additive bias, ϵ follows normal distribution with zero
 486 mean, τ standard deviation with no correlation within or between players. Player
 487 A's prediction and revised prediction can be written as follows:

$$Y_A = c_A X_{e_A} \quad (\text{S17})$$

488

$$Y_A^{re} = \beta_A Y_A^{AI} + (1 - \beta_A) Y_A \quad (\text{S18})$$

489 where c_A stands for player A's belief on the factor, e_A stands for the interest, and
 490 β_A stands for reliance on AI.

491 2.2 Aggregation

492 For Chatbot AI, clustering aggregation is used for collective prediction.

$$\begin{aligned} \hat{Y}^{re} &= \sum_{i=0}^M \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} Y_A^{re} \\ &= \sum_{i=0}^M \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \{\beta_A Y_A^{AI} + (1 - \beta_A) Y_A\} \\ &= \sum_{i=0}^M \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} [\{\beta_A \alpha_{e_A} + (1 - \beta_A) c_A\} X_{e_A} + \beta_A b_{e_A} + \beta_A \epsilon] \end{aligned}$$

493 Reparameterize for the simplicity.

$$\tilde{c}_i := \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \{\beta_A \alpha_{e_A} + (1 - \beta_A) c_A\} \quad (\text{S19})$$

494

$$\tilde{B}_i := \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \beta_A b_{e_A} \quad (\text{S20})$$

495

$$\bar{B} := \sum_{i=0}^M \tilde{B}_i \quad (\text{S21})$$

496

$$\tilde{\eta}_i := \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \beta_A \epsilon \quad (\text{S22})$$

497

$$\bar{\eta} := \sum_{i=0}^M \tilde{\eta}_i \quad (\text{S23})$$

498 2.3 Expected payoffs

499 Feedback

500

$$\delta_0 := \alpha_0 - \tilde{c}_0 - \bar{B} \quad (\text{S24})$$

501

$$\delta_i := \alpha_i - \tilde{c}_i, \quad (i \neq 0) \quad (\text{S25})$$

$$Y - \hat{Y}^{re} = \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta} \quad (\text{S26})$$

502 δ_0 is constant term and δ_i is independent of α_k ($k \neq i$), ϵ , and $\bar{\eta}$.

$$\begin{aligned} \mathbb{E}(\pi_A) &= \mathbb{E}[Y_A^{re}(Y - \hat{Y}^{re})] \\ &= \mathbb{E} \left[\{(\beta_A \alpha_{e_A} + (1 - \beta_A) c_A) X_{e_A} + \beta_A b_{e_A} + \beta_A \epsilon\} \left\{ \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta} \right\} \right] \\ &= \{\beta_A \alpha_{e_A} + (1 - \beta_A) c_A\} \delta_{e_A} \sigma_{e_A}^2 + \beta_A b_{e_A} \delta_0 - \frac{1}{n_{e_A}} \beta_A^2 \tau^2 \end{aligned}$$

503 Niche-expert

504 If $e_A = 0$,

$$\begin{aligned} \mathbb{E}(\pi_A) &= -\mathbb{E}[\rho_0(Y_A^{re} - Y)^2] \\ &= -\mathbb{E} \left[\rho_0 \left[\{\beta_A \alpha_0 + (1 - \beta_A) c_A\} + \beta_A b_0 + \beta_A \epsilon - \alpha_0 - \sum_{i=1}^M \alpha_i X_i \right]^2 \right] \\ &= -\rho_0 \left\{ (1 - \beta_A)^2 (c_A - \alpha_0)^2 + 2(1 - \beta_A)(c_A - \alpha_0) \beta_A b_0 + \beta_A^2 b_0^2 \right. \\ &\quad \left. + \sum_{i \neq 0} \alpha_i^2 \sigma_i^2 + \beta_A^2 \tau^2 \right\} \end{aligned}$$

505 Else,

$$\begin{aligned}
\mathbb{E}(\pi_A) &= -\mathbb{E}[\rho_{e_A}(Y_A^{\text{re}} - Y)^2] \\
&= -\mathbb{E}\left[\rho_{e_A}\left[\{\beta_A\alpha_{e_A} + (1 - \beta_A)c_A\}X_{e_A} + \beta_A b_{e_A} + \beta_A\epsilon - \alpha_0 - \sum_{i=1}^M \alpha_i X_i\right]^2\right] \\
&= -\rho_{e_A}\left\{(1 - \beta_A)^2(c_A - \alpha_{e_A})^2\sigma_{e_A}^2 + \sum_{i \neq e_A, 0} \alpha_i^2 \sigma_i^2\right. \\
&\quad \left.+ \alpha_0^2 + \beta_A^2 b_{e_A}^2 - 2\alpha_0\beta_A b_{e_A} + \beta_A^2 \tau^2\right\}
\end{aligned}$$

506 Incentivizing or penalizing AI use

507 Because additive term $\lambda\beta$ is not a random variable, we can simply derive the expected
508 payoff as

$$\mathbb{E}(\pi_{A, \text{Feedback}}) = \mathbb{E}\left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) + \lambda\beta_A\right] = \mathbb{E}\left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}})\right] + \lambda\beta_A. \quad (\text{S27})$$

$$\mathbb{E}(\pi_{A, \text{Niche-expert}}) = -\mathbb{E}\left[\rho_{e_A}(Y_A^{\text{re}} - Y)^2 + \lambda\beta_A\right] = -\mathbb{E}\left[\rho_{e_A}(Y_A^{\text{re}} - Y)^2\right] + \lambda\beta_A. \quad (\text{S28})$$

509 Balanced

510 Expected payoff of the balanced structure can also be driven by arithmetic mean of
511 expected payoff of feedback and niche-expert.

$$\mathbb{E}(\pi_A) = \mathbb{E}\left[\frac{Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) - \rho_{e_A}(Y_A^{\text{re}} - Y)^2}{2}\right] = \frac{\mathbb{E}\left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}})\right] - \mathbb{E}\left[\rho_{e_A}(Y_A^{\text{re}} - Y)^2\right]}{2} \quad (\text{S29})$$

512 2.4 Expected accuracy

$$\mathbb{E}[(Y - \alpha_0)^2] = \mathbb{E}\left[\left(\alpha_0 + \sum_{i=1}^M \alpha_i X_i - \alpha_0\right)^2\right] = \sum_{i=1}^M \alpha_i^2 \sigma_i^2 \quad (\text{S30})$$

$$\mathbb{E}[(Y - \hat{Y})^2] = \mathbb{E}\left[\left(\delta_0 + \sum_{i=1}^M \delta_i X_i - \hat{\eta}\right)^2\right] = \delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \sum_{i=0}^M \frac{\tau^2}{n_i^2} \sum_{A \in \mathcal{A}_i} \beta_A^2 \quad (\text{S31})$$

$$\begin{aligned}
R &= 1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]} \\
&= 1 - \frac{\delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \sum_{i=0}^M \frac{\tau^2}{n_i^2} \sum_{A \in \mathcal{A}_i} \beta_A^2}{\sum_{i=1}^M \alpha_i^2 \sigma_i^2}
\end{aligned}$$

515 3 Evolutionary dynamics

516 For sufficiently large population ($N \rightarrow \infty$), we can approximate the deterministic
517 evolutionary dynamics using the replicator equation. Let λ denote a strategy, p_λ its
518 frequency, and $u_\lambda = \mathbb{E}(\pi_\lambda)$ its expected payoff. The replicator equation is

$$\frac{dp_\lambda}{dt} = p_\lambda(u_\lambda - \bar{u}), \quad (\text{S32})$$

519 where $\bar{u} = \sum_\lambda p_\lambda u_\lambda$. Therefore, at an internal stationary state, all strategies
520 with positive frequency have the same expected payoff. A boundary state can also
521 be stationary because a strategy with exactly zero frequency cannot increase under
522 the replicator equation. Thus, stability of a boundary state needs to be considered
523 separately by examining invasion by a rare strategy.

524 3.1 Omniscient AI

525 3.1.1 Feedback

526 When the population size approaches infinity, the effect of stochastic error on the
527 collective prediction vanishes ($\bar{\eta} \rightarrow 0$) and the effect of stochastic error in the ex-
528 pected payoff, $-\beta^2 \tau^2 / N$, also vanishes. Following prior research [13], we define the
529 Lyapunov function as the expected error of collective prediction

$$V = \mathbb{E} \left[(Y - \hat{Y}^{re})^2 \right]. \quad (\text{S33})$$

530 The function V is non-negative and equals zero when the collective prediction is
531 identical to the target. To examine how V changes under the evolutionary dynamics,
532 we take its time derivative:

$$\begin{aligned}
\frac{dV}{dt} &= -2\mathbb{E} \left[(Y - \hat{Y}^{re}) \frac{d\hat{Y}^{re}}{dt} \right] \\
&= -2\mathbb{E} \left[\left(\delta_0 + \sum_{i=1}^M \delta_i X_i \right) \left\{ \frac{d\bar{\beta}}{dt} (Y + B) + \sum_{i=0}^M \frac{d\gamma_i}{dt} X_i \right\} \right] \\
&= -2 \left[\left\{ (\alpha_0 + B) \delta_0 + \sum_{i=1}^M \alpha_i \delta_i \sigma_i^2 \right\} \frac{d\bar{\beta}}{dt} + \delta_0 \frac{d\gamma_0}{dt} + \sum_{i=1}^M \delta_i \sigma_i^2 \frac{d\gamma_i}{dt} \right].
\end{aligned}$$

533 For notational convenience, we define

$$K = (\alpha_0 + B) \delta_0 + \sum_{i=1}^M \alpha_i \delta_i \sigma_i^2. \quad (\text{S34})$$

534 Although K changes with the population state, it is common to all strategies at a
535 given state and can therefore be treated as a constant with respect to the strategy
536 index λ . We also define

$$G_e = \begin{cases} \delta_0, & e = 0, \\ \delta_e \sigma_e^2, & e \neq 0. \end{cases} \quad (\text{S35})$$

537 By the definitions of $\bar{\beta}$ and γ_i and the replicator equation,

$$\frac{d\bar{\beta}}{dt} = \sum_{\lambda} p_{\lambda} (u_{\lambda} - \bar{u}) \beta_{\lambda} = \text{Cov}(\beta, u). \quad (\text{S36})$$

538 Similarly, the terms involving γ_i can be combined as

$$\delta_0 \frac{d\gamma_0}{dt} + \sum_{i=1}^M \delta_i \sigma_i^2 \frac{d\gamma_i}{dt} = \text{Cov}((1 - \beta) c G_e, u). \quad (\text{S37})$$

539 Therefore,

$$\begin{aligned}
\frac{dV}{dt} &= -2 \{ K \text{Cov}(\beta, u) + \text{Cov}((1 - \beta) c G_e, u) \} \\
&= -2 \text{Cov}(K\beta + (1 - \beta) c G_e, u).
\end{aligned}$$

540 For the feedback structure in the limit $N \rightarrow \infty$, the expected payoff of strategy
541 λ is

$$u_{\lambda} = K \beta_{\lambda} + (1 - \beta_{\lambda}) c_{\lambda} G_{e_{\lambda}}. \quad (\text{S38})$$

542 Which is same with Thus,

$$\frac{dV}{dt} = -2 \text{Cov}(u, u) = -2 \text{Var}(u) \leq 0. \quad (\text{S39})$$

Therefore, the expected error of collective prediction cannot increase during the evolutionary process and decreases whenever strategies differ in their expected payoffs. The derivative becomes zero when the variance of expected payoffs is zero, $\text{Var}(u) = 0$. This condition corresponds to equal expected payoffs among the strategies present in the population. A boundary state in which the entire population consists of a single strategy can also satisfy $\text{Var}(u) = 0$ even when $V > 0$. However, such a state is not evolutionarily stable under the feedback structure. If the collective prediction differs from the target Y , a nonzero residual remains. By the definition of the feedback payoff, a novel strategy whose prediction aligns with the direction of this residual obtains a higher expected payoff than the resident strategy and can therefore invade the population. Thus, an imperfect boundary equilibrium with $V > 0$ is unstable to the introduction of novel predictions.

Consequently, the evolutionary dynamics continue to reduce the collective prediction error until the population reaches the stable state with zero residual, $V = 0$. At this state, the collective prediction equals the target, $\hat{Y}^{re} = Y$, and the collective accuracy reaches $R = 1$.

3.1.2 Niche-expert

Unlike the stochastic error in the collective prediction, stochastic error in the niche-expert payoff is evaluated at the individual level and does not vanish as population size increases. Therefore, we retain $\tau > 0$ in the following derivation.

We first consider strategies having the same interest by optimizing the expected payoff with respect to c and β . For a nonzero interest e , the expected payoff is strictly concave with respect to c for $\beta < 1$ and $\sigma_e > 0$, and is maximized at

$$c_e^* = \alpha_e. \quad (\text{S40})$$

After substituting $c_e^* = \alpha_e$, the payoff is strictly concave with respect to β . Its unconstrained stationary point is

$$\tilde{\beta}_e = \frac{\alpha_0^2 + \alpha_0 B + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2}{(\alpha_0 + B)^2 + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S41})$$

Because β is constrained between 0 and 1, the optimal β_e^* is $\tilde{\beta}_e$ when $0 < \tilde{\beta}_e < 1$, and otherwise lies at the corresponding boundary, $\beta_e^* = 0$ or $\beta_e^* = 1$. Thus, the boundary cases do not require a separate evolutionary argument; they only change the optimal prediction and its minimum expected squared error. When $\beta_e^* = 1$, the revised prediction is independent of c_e , so different values of c_e are selectively neutral and can be treated as a single payoff-equivalent strategy class.

Let L_e denote the minimum expected individual error, $\mathbb{E}[(Y_A^{re} - Y)^2]$, within interest e . For an internal optimum,

$$L_e = \frac{B^2 \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 + \tau^2 \left(\alpha_0^2 + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 \right)}{(\alpha_0 + B)^2 + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S42})$$

576 If the optimum is located at $\beta_e^* = 0$ or $\beta_e^* = 1$, L_e is instead obtained by evaluating
 577 the expected squared prediction error at the corresponding boundary.

578 For interest 0, optimization with respect to c gives

$$c_0^*(\beta) = \alpha_0 - \frac{\beta}{1 - \beta} B. \quad (\text{S43})$$

579 Substituting this into the expected payoff gives a strictly concave function of β ,
 580 with the unique optimum

$$\beta_0^* = \frac{\sum_{j=1}^M \alpha_j^2 \sigma_j^2}{\sum_{j=1}^M \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S44})$$

581 Therefore,

$$c_0^* = \alpha_0 - \frac{B}{\tau^2} \sum_{j=1}^M \alpha_j^2 \sigma_j^2, \quad (\text{S45})$$

582 and

$$L_0 = \frac{\tau^2 \sum_{j=1}^M \alpha_j^2 \sigma_j^2}{\sum_{j=1}^M \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S46})$$

583 We denote the optimal strategy, or the payoff-equivalent strategy class when
 584 $\beta_e^* = 1$, by S_e . A state containing a suboptimal strategy within an interest can
 585 be stationary if the optimal strategy is absent, but it is unstable against invasion
 586 because the optimal strategy has a strictly higher payoff at the same ρ_e . Therefore,
 587 a stable state can contain only S_e within each represented interest.

588 We next consider the frequencies of the $M + 1$ optimal strategies. Let ρ_e denote
 589 the frequency of S_e . Its expected payoff is

$$u_e = -\rho_e L_e. \quad (\text{S47})$$

590 Assuming $L_e > 0$ for all interests, the equal-payoff condition of the replicator
 591 equation at an internal stationary state gives

$$\rho_0^* L_0 = \rho_1^* L_1 = \dots = \rho_M^* L_M. \quad (\text{S48})$$

592 Together with $\sum_{e=0}^M \rho_e^* = 1$, this uniquely determines

$$\rho_e^* = \frac{D}{L_e}, \quad D = \left(\sum_{j=0}^M \frac{1}{L_j} \right)^{-1}. \quad (\text{S49})$$

593 This internal stationary state is locally stable. At equilibrium, all represented
 594 interests have the same expected payoff, with $\rho_e^* L_e = D$. If the frequency of one
 595 interest increases slightly above its equilibrium value, its payoff decreases because
 596 the niche-expert payoff is proportional to $-\rho_e L_e$. Conversely, if the frequency of
 597 an interest decreases below its equilibrium value, its payoff increases. The replicator
 598 dynamics therefore generate a restoring force that reduces small deviations in interest
 599 frequencies and drives the population back toward the internal stationary state.

600 If one or more interests are absent, the resulting boundary state can be stationary
 601 under the replicator equation, but it is unstable against invasion. The payoff of a rare
 602 absent interest approaches zero as $\rho_e \rightarrow 0$, whereas the mean payoff of the resident
 603 population is negative when $L_e > 0$, so the rare interest increases when introduced
 604 at sufficiently small frequency.

605 Thus, there exists a unique stable full-support equilibrium in terms of the optimal
 606 prediction strategies and their frequencies. Whether the constrained optimum has
 607 $\beta_e^* = 0$, $0 < \beta_e^* < 1$, or $\beta_e^* = 1$ does not change the frequency argument, because
 608 these cases differ only through the resulting value of L_e . When $\beta_e^* = 1$, the strategy
 609 is selectively neutral with respect to c_e , but its prediction, payoff, and equilibrium
 610 interest frequency remain uniquely determined.

611 This equilibrium also allows a direct comparison with the expected accuracy of
 612 the AI. Using Y^{AI} itself without aggregation as the prediction gives

$$\mathbb{E} [(Y - Y^{\text{AI}})^2] = B^2 + \tau^2. \quad (\text{S50})$$

613 For every interest, complete reliance on AI, $\beta = 1$, is a feasible strategy. There-
 614 fore, because L_e is the minimum individual expected squared prediction error within
 615 interest e ,

$$L_e \leq B^2 + \tau^2. \quad (\text{S51})$$

616 Moreover, for interest 0,

$$L_0 = \frac{\tau^2 \sum_{j=1}^M \alpha_j^2 \sigma_j^2}{\sum_{j=1}^M \alpha_j^2 \sigma_j^2 + \tau^2} < \tau^2 \leq B^2 + \tau^2. \quad (\text{S52})$$

617 Because the stable equilibrium has $\rho_0^* > 0$,

$$\sum_{e=0}^M \rho_e^* L_e < B^2 + \tau^2. \quad (\text{S53})$$

618 The collective prediction is the average of individual revised predictions. By
 619 convexity of the squared error,

$$\mathbb{E} [(Y - \hat{Y}^{re})^2] \leq \sum_{e=0}^M \rho_e^* L_e. \quad (\text{S54})$$

620 Therefore,

$$\mathbb{E}[(Y - \hat{Y}^{re})^2] < \mathbb{E}[(Y - Y^{AI})^2]. \quad (\text{S55})$$

621 Because the collective accuracy and AI accuracy use the same normalization by
 622 $\mathbb{E}[(Y - \alpha_0)^2]$, the stationary collective accuracy under the niche-expert structure is
 623 strictly greater than the expected accuracy of the AI prediction.

624 Although the niche-expert structure improves collective prediction relative to the
 625 AI, the stationary collective accuracy does not reach one when $\tau > 0$. For a nonzero
 626 factor i , the residual coefficient of the stationary collective prediction can be written
 627 as

$$\delta_i^* = \alpha_i \{1 - \bar{\beta}^* - \rho_i^*(1 - \beta_i^*)\} = \alpha_i \sum_{e \neq i} \rho_e^*(1 - \beta_e^*). \quad (\text{S56})$$

628 At the stable equilibrium, $\rho_0^* > 0$. In addition, $\tau > 0$ implies $\beta_0^* < 1$. Therefore,
 629 for every factor with $\alpha_i \neq 0$,

$$\delta_i^* \neq 0. \quad (\text{S57})$$

630 Because $\sum_{i=1}^M \alpha_i^2 \sigma_i^2 > 0$, at least one factor contributes positive prediction error.
 631 Hence,

$$\mathbb{E}[(Y - \hat{Y}^{re})^2] > 0, \quad (\text{S58})$$

632 and consequently the stationary collective accuracy satisfies $R^* < 1$.

633 3.2 Chatbot AI

634 3.2.1 Feedback

635 With Chatbot AI, the effect of stochastic error on the collective prediction depends
 636 on n_i , requiring the additional assumption that $n_i \rightarrow \infty$ for every interest for the
 637 stochastic error to vanish. However, Under this assumption, the loss of interest
 638 diversity cannot occur, and therefore the stochastic breakdown of collective accuracy
 639 observed in finite populations cannot be analytically captured. Instead, by assuming
 640 that each interest is represented by infinitely many players, we can show that the
 641 expected collective prediction error monotonically decreases until perfect accuracy
 642 is achieved. Because finite population dynamics are more relevant to the empirical
 643 populations considered in this study, we retain finite population simulations for the
 644 main result.

645 3.2.2 Niche-expert

646 We first consider strategies having the same interest by optimizing the expected
 647 payoff with respect to c and β . For a nonzero interest e , the expected payoff is
 648 strictly concave with respect to c for $\beta < 1$ and $\sigma_e > 0$, and is maximized at

$$c_e^* = \alpha_e. \quad (\text{S59})$$

649 After substituting $c_e^* = \alpha_e$, the payoff is strictly concave with respect to β . Its
 650 unconstrained stationary point is

$$\tilde{\beta}_e = \frac{2\alpha_0 b_e}{2b_e^2 + 2\tau^2}. \quad (\text{S60})$$

651 Because β is constrained between 0 and 1, the optimal β_e^* is $\tilde{\beta}_e$ when $0 < \tilde{\beta}_e < 1$,
 652 and otherwise lies at the corresponding boundary, $\beta_e^* = 0$ or $\beta_e^* = 1$. Because we
 653 assumes AI model with moderate accuracy, b_e and τ is relatively smaller than α_0
 654 which makes $\tilde{\beta}_e > 1$ when $\alpha_0 b_e > 0$ and $\tilde{\beta}_e < 0$ when $\alpha_0 b_e < 0$. These boundary
 655 cases lead to extreme reliance on AI or complete exclusion of AI which both hinders
 656 the emergence of CI. Only when the prediction task is centered near zero and AI is
 657 relatively inaccurate ($\alpha_0 b_e < b_e^2 + \tau^2$), and $\alpha_0 b_e > 0$, the internal optimum β_e^* can be
 658 exist.

659 For interest 0, optimization with respect to c gives

$$c_0^*(\beta) = \alpha_0 - \frac{\beta}{1 - \beta} b_0. \quad (\text{S61})$$

660 Substituting this into the expected payoff gives a strictly concave function of β , with
 661 the unique optimum

$$\beta_0^* = 0. \quad (\text{S62})$$

662 **Supplementary Figure**

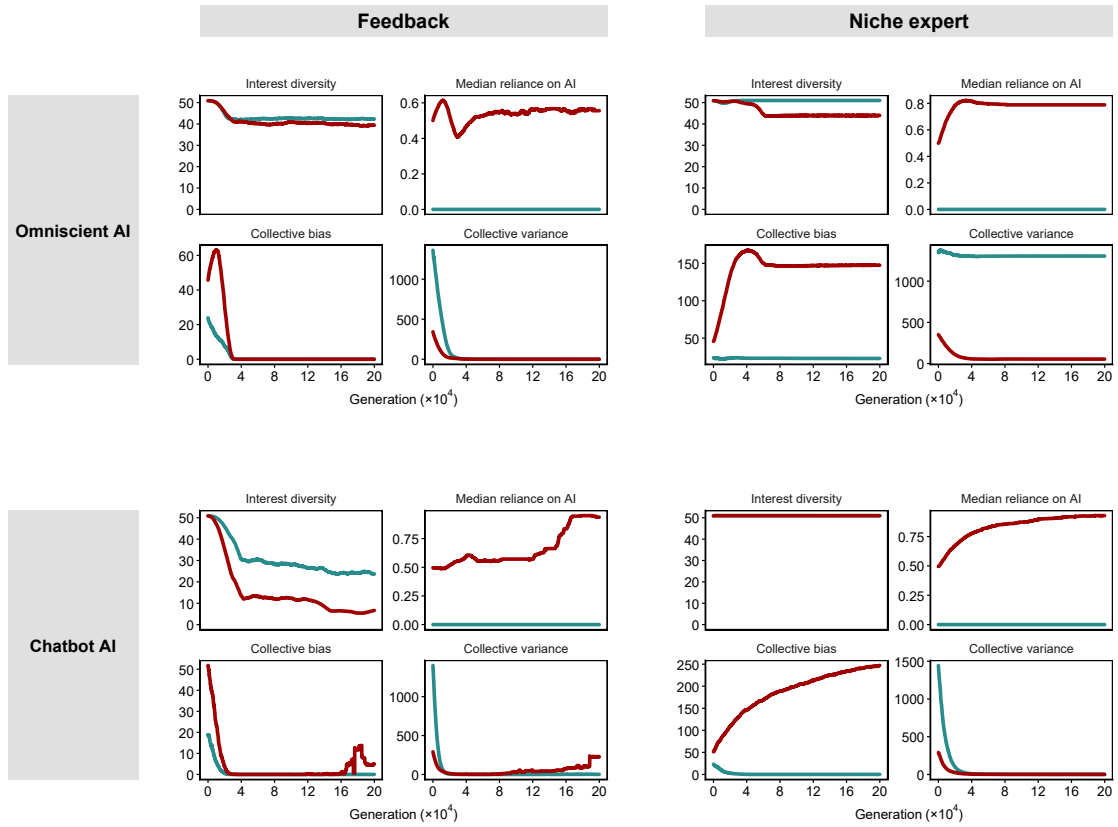


Figure S1: **Caption.** Text.

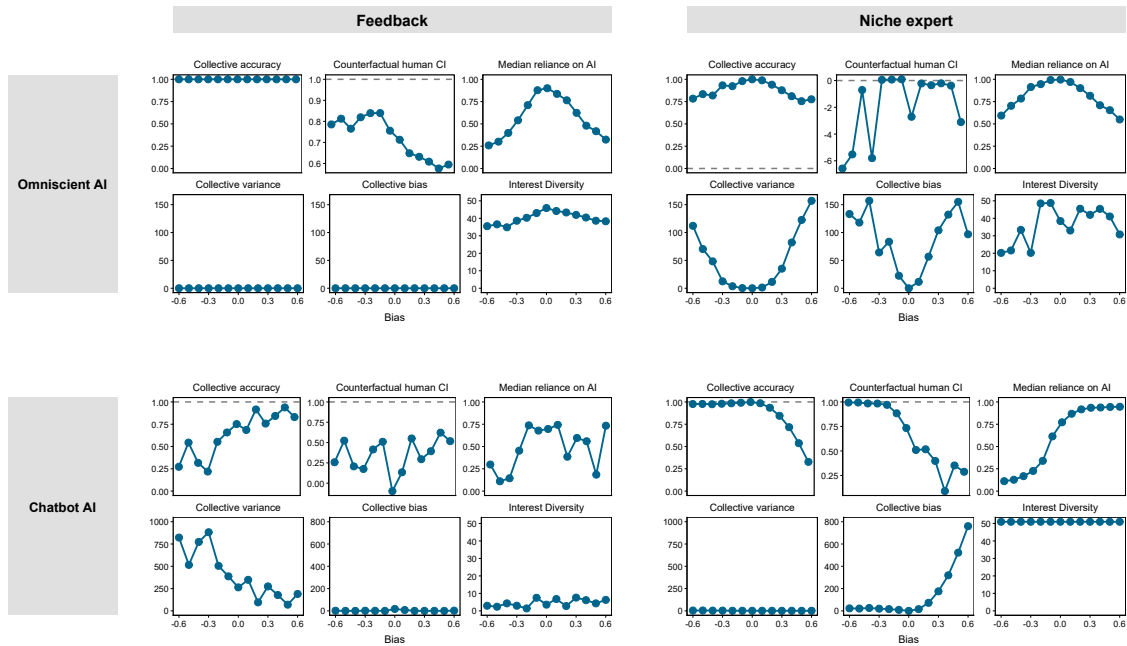


Figure S2: Caption. Text.

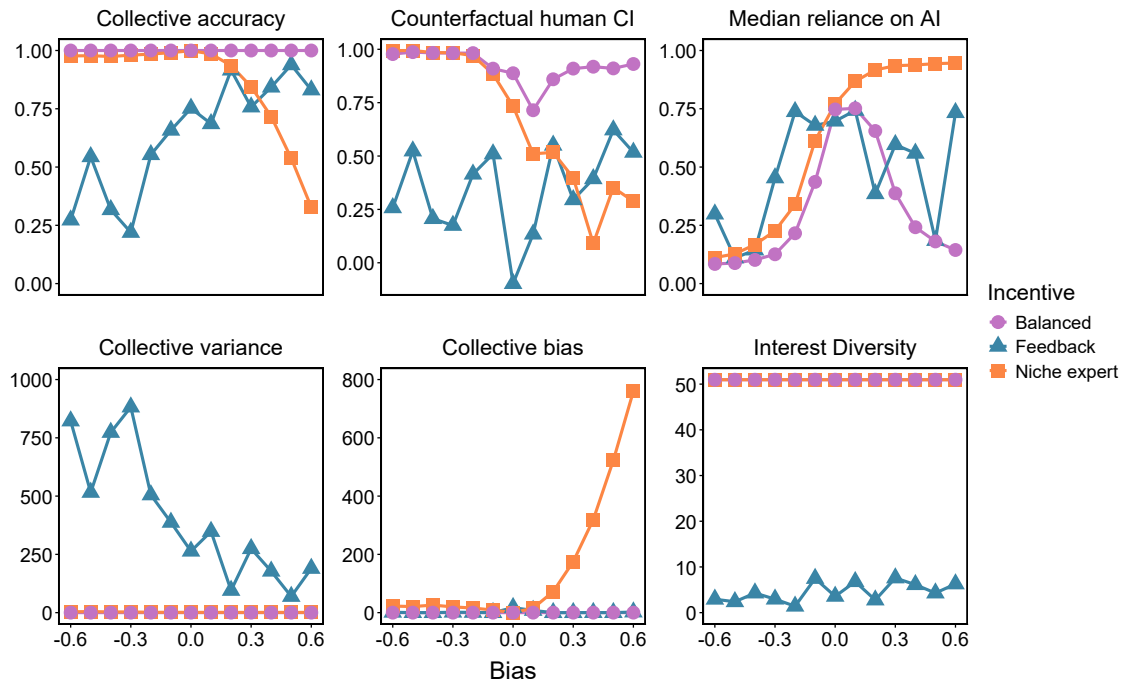


Figure S3: **Caption.** Text.

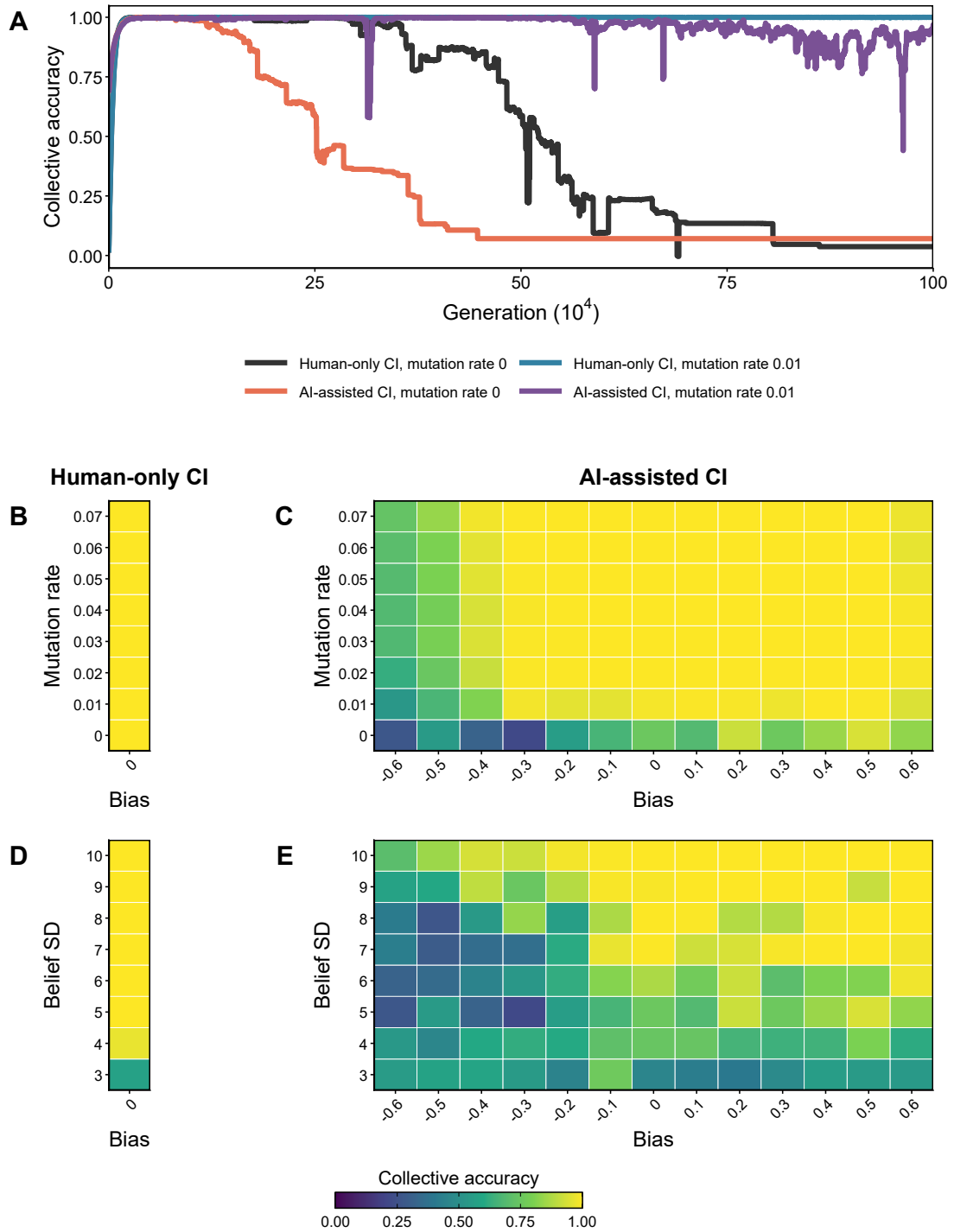


Figure S4: **Caption.** Text.

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